In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
df = pd.read_csv('US_honey_dataset.csv')
```

In [3]:
```
df.shape
```

Out[3]:
```
(1115, 9)
```

In [4]:
```
df.head()
```

Out[4]:

	Unnamed: 0	state	colonies_number	yield_per_colony	production	stocks	average_price	value_of_production	year
0	0	Alabama	16000	58	928000	28000	62.0	575000	1995
1	1	Arizona	52000	79	4108000	986000	68.0	2793000	1995
2	2	Arkansas	50000	60	3000000	900000	64.0	1920000	1995
3	3	California	420000	93	39060000	4687000	60.0	23436000	1995
4	4	Colorado	45000	60	2700000	1404000	68.0	1836000	1995

In [5]:
```
df.tail()
```

Out[5]:

	Unnamed: 0	state	colonies_number	yield_per_colony	production	stocks	average_price	value_of_production	year
1110	1110	Virginia	6000	40	79000	79000	8.23	1975000	2021
1111	1111	Washington	96000	32	1206000	1206000	2.52	7741000	2021
1112	1112	WestVirginia	6000	43	136000	136000	4.80	1238000	2021
1113	1113	Wisconsin	42000	47	750000	750000	2.81	5547000	2021
1114	1114	Wyoming	38000	58	242000	242000	2.07	4562000	2021

In [6]:
```
df.columns
```

Out[6]:
```
Index(['Unnamed: 0', 'state', 'colonies_number', 'yield_per_colony',
       'production', 'stocks', 'average_price', 'value_of_production', 'year'],
      dtype='object')
```

In [7]:
```
df.describe()
```

Out[7]:

	Unnamed: 0	colonies_number	yield_per_colony	production	stocks	average_price	value_of_production	year
count	1115.00000	1115.000000	1115.000000	1.115000e+03	1.115000e+03	1115.000000	1.115000e+03	1115.000000
mean	557.00000	62438.565022	59.743498	2.851268e+06	1.172625e+06	140.623076	5.667412e+06	2007.740807
std	322.01708	92648.175955	19.940500	5.561202e+06	2.049556e+06	107.011544	9.459460e+06	7.823002
min	0.00000	2000.000000	19.000000	1.200000e+04	9.000000e+03	1.300000	1.060000e+05	1995.000000
25%	278.50000	9000.000000	45.000000	2.460000e+05	1.125000e+05	70.000000	1.008000e+06	2001.000000
50%	557.00000	26000.000000	57.000000	8.280000e+05	3.700000e+05	128.000000	2.281000e+06	2008.000000
75%	835.50000	69000.000000	71.000000	2.700000e+06	1.253500e+06	193.000000	5.704000e+06	2015.000000
max	1114.00000	550000.000000	155.000000	3.906000e+07	1.354500e+07	874.000000	8.385900e+07	2021.000000

In [8]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1115 entries, 0 to 1114
Data columns (total 9 columns):
 #   Column               Non-Null Count  Dtype  
---  ------               --------------  -----  
 0   Unnamed: 0           1115 non-null   int64  
 1   state                1115 non-null   object 
 2   colonies_number      1115 non-null   int64  
 3   yield_per_colony     1115 non-null   int64  
 4   production           1115 non-null   int64  
 5   stocks               1115 non-null   int64  
 6   average_price        1115 non-null   float64
 7   value_of_production  1115 non-null   int64  
 8   year                 1115 non-null   int64  
dtypes: float64(1), int64(7), object(1)
memory usage: 78.5+ KB
```

In [9]:
```
df.isnull().sum()
```

Out[9]:
```
Unnamed: 0            0
state                 0
colonies_number       0
yield_per_colony      0
production            0
stocks                0
average_price         0
value_of_production   0
year                  0
dtype: int64
```

In [10]:
```
df.duplicated().sum()
```

Out[10]:
```
np.int64(0)
```

In [11]:
```
df.drop(columns =['Unnamed: 0'] ,inplace =True)
```

In [12]:
```
df.columns
```

Out[12]:
```
Index(['state', 'colonies_number', 'yield_per_colony', 'production', 'stocks',
       'average_price', 'value_of_production', 'year'],
      dtype='object')
```

In [13]:
```
df.describe()
```

Out[13]:

	colonies_number	yield_per_colony	production	stocks	average_price	value_of_production	year
count	1115.000000	1115.000000	1.115000e+03	1.115000e+03	1115.000000	1.115000e+03	1115.000000
mean	62438.565022	59.743498	2.851268e+06	1.172625e+06	140.623076	5.667412e+06	2007.740807
std	92648.175955	19.940500	5.561202e+06	2.049556e+06	107.011544	9.459460e+06	7.823002
min	2000.000000	19.000000	1.200000e+04	9.000000e+03	1.300000	1.060000e+05	1995.000000
25%	9000.000000	45.000000	2.460000e+05	1.125000e+05	70.000000	1.008000e+06	2001.000000
50%	26000.000000	57.000000	8.280000e+05	3.700000e+05	128.000000	2.281000e+06	2008.000000
75%	69000.000000	71.000000	2.700000e+06	1.253500e+06	193.000000	5.704000e+06	2015.000000
max	550000.000000	155.000000	3.906000e+07	1.354500e+07	874.000000	8.385900e+07	2021.000000

In [14]:
```
df['state'].value_counts().sort_index()
```

Out[14]:

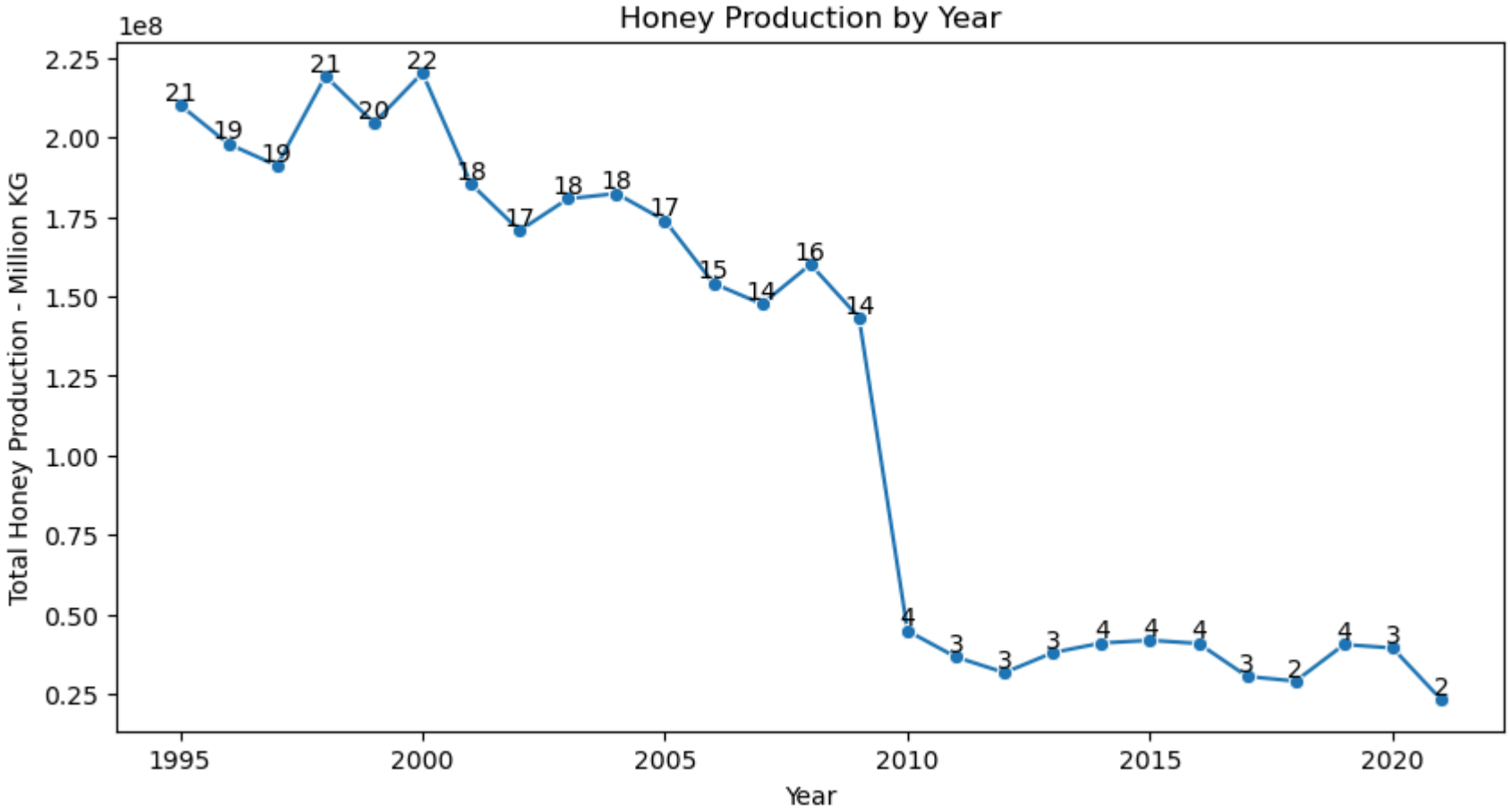
state	
Alabama	27
Arizona	27
Arkansas	27
California	27
Colorado	27
Florida	27
Georgia	27
Hawaii	27
Idaho	27
Illinois	27
Indiana	27
Iowa	27
Kansas	27
Kentucky	26
Louisiana	27
Maine	27
Maryland	9
Michigan	27
Minnesota	27
Mississippi	27
Missouri	27
Montana	27
Nebraska	27
Nevada	15
NewJersey	27
NewMexico	18
NewYork	27
NorthCarolina	27
NorthDakota	27
Ohio	27
Oklahoma	9
Oregon	27
Pennsylvania	27
SouthCarolina	12
SouthDakota	27
Tennessee	27
Texas	27
Utah	27
Vermont	27

```
Virginia          27
Washington        27
WestVirginia      27
Wisconsin         27
Wyoming           27
Name: count, dtype: int64
```

In [15]:

```python
year_production = df.groupby("year")["production"].sum()
print(year_production)
plt.figure(figsize=(10, 5))
sns.lineplot(x=year_production.index, y=year_production.values, marker="o")
for x, y in zip(year_production.index, year_production.values):
    plt.text(x, y, f"{int(y / 1_000_000) // 10}", ha="center", va="bottom")
plt.title("Honey Production by Year")
plt.xlabel("Year")
plt.ylabel("Total Honey Production - Million KG")
plt.show()
```

```
year
1995    210272000
1996    197995000
1997    191176000
1998    219321000
1999    204585000
2000    220320000
2001    185403000
2002    170732000
2003    180741000
2004    182379000
2005    173857000
2006    154120000
2007    147609000
2008    160096000
2009    143340000
2010     44797000
2011     36550000
2012     31662000
2013     37974000
2014     40990000
2015     41832000
2016     40837000
2017     30478000
2018     28989000
2019     40510000
2020     39412000
2021     23187000
Name: production, dtype: int64
```



In [16]:
```
year_revenue = df.groupby("year")["value_of_production"].sum()

highest_year = year_revenue.idxmax()
lowest_year = year_revenue.idxmin()

colors = []

for year in year_revenue.index:
    if year == highest_year:
        colors.append("green")
    elif year == lowest_year:
        colors.append("red")
```

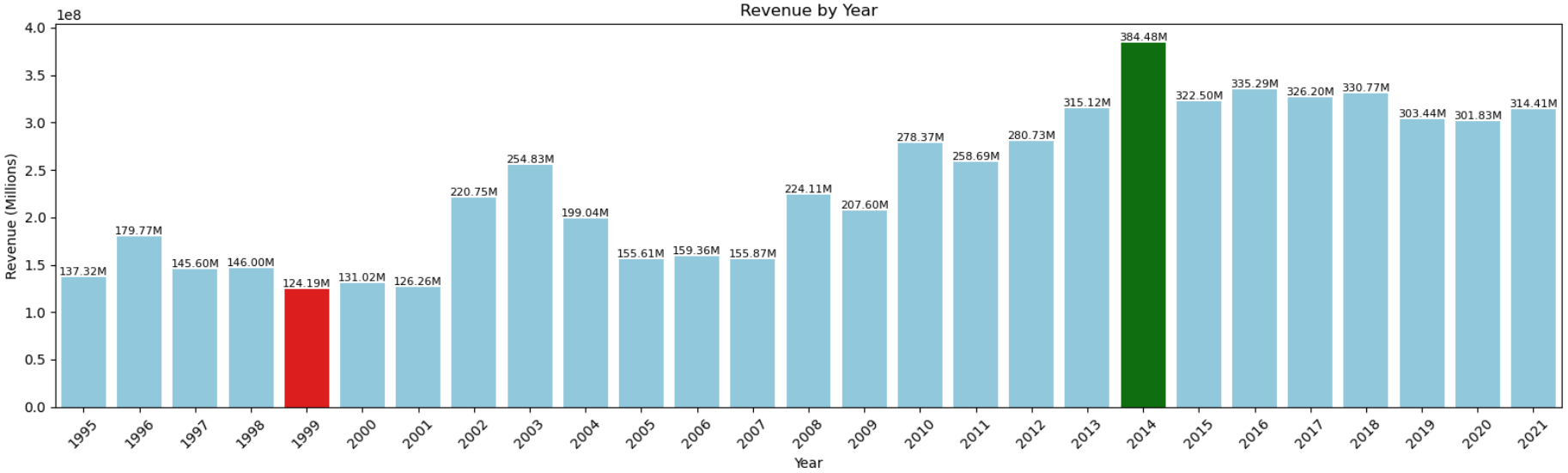
```
    else:
        colors.append("skyblue")

plt.figure(figsize=(16, 5))

ax = sns.barplot(
    x=year_revenue.index,
    y=year_revenue.values,
    palette=colors,
    hue=year_revenue.index,
    legend=False
)

for bar in ax.patches:
    height = bar.get_height()
    ax.annotate(
        f"{height / 1_000_000:.2f}M",
        (bar.get_x() + bar.get_width() / 2, height),
        ha="center",
        va="bottom",
        fontsize=8
    )

plt.title("Revenue by Year")
plt.xlabel("Year")
plt.ylabel("Revenue (Millions)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



In [17]:
```
avg_price_year = df.groupby("year")["average_price"].mean()
print(avg_price_year)
```

```
year
1995     74.840909
1996     99.568182
1997     91.325581
1998     83.720930
1999     80.325581
2000     79.023256
2001     88.465116
2002    133.204545
2003    151.068182
2004    132.350000
2005    116.341463
2006    134.341463
2007    140.170732
2008    160.878049
2009    176.195122
2010    192.900000
2011    216.725000
2012    236.700000
2013    260.717949
2014    282.025000
2015    292.625000
2016    291.925000
2017    302.775000
2018      3.083500
2019      2.925500
2020      3.050000
2021      3.334250
Name: average_price, dtype: float64
```

In [18]:
```
avg_price_state = df.groupby(["year", "state"])["average_price"].mean()
print(avg_price_state)
```

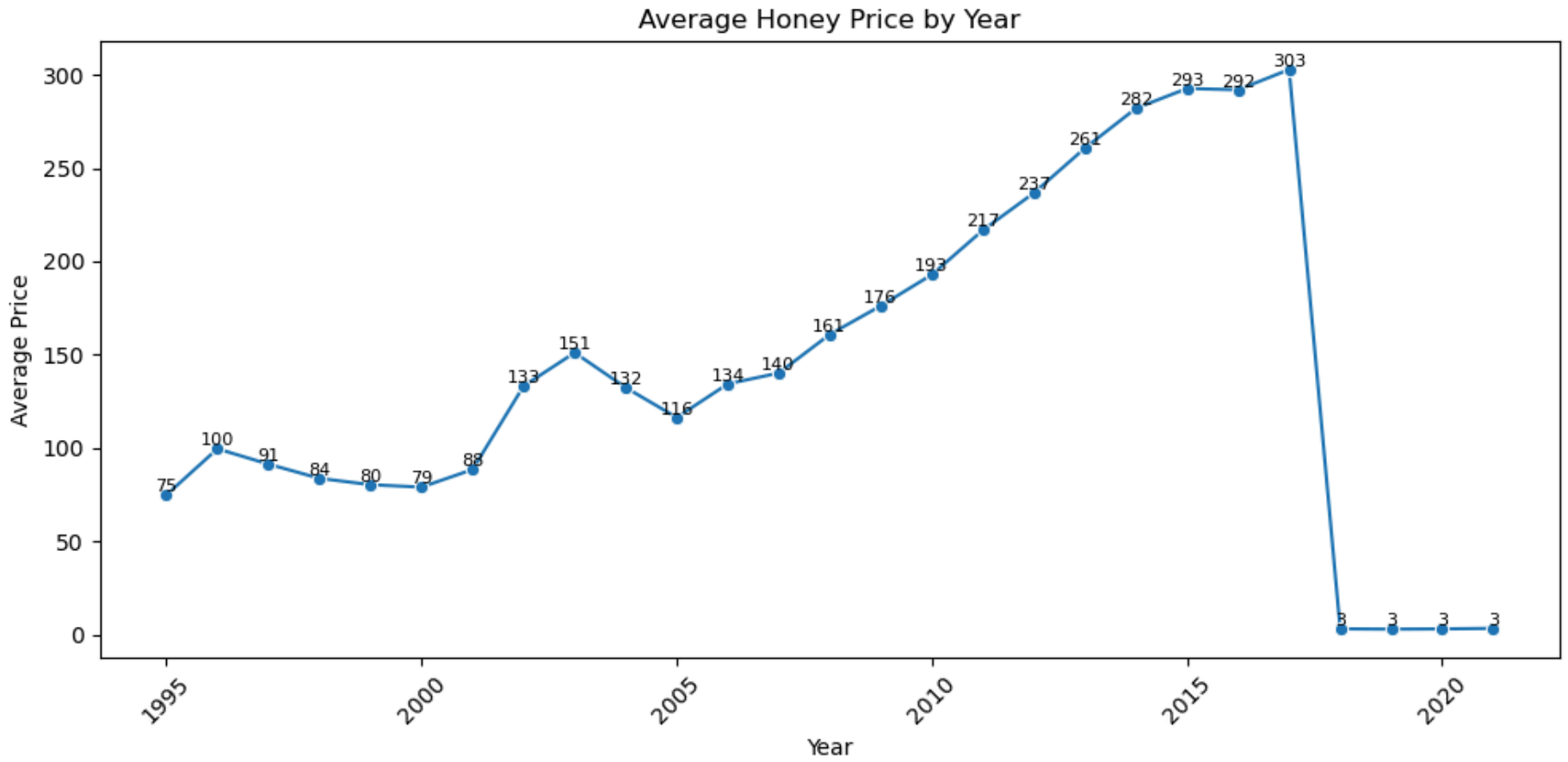
```
year  state
1995  Alabama         62.00
      Arizona         68.00
      Arkansas        64.00
      California      60.00
      Colorado        68.00
                      ...
2021  Virginia         8.23
      Washington       2.52
      WestVirginia     4.80
      Wisconsin        2.81
      Wyoming          2.07
Name: average_price, Length: 1115, dtype: float64
```

In [19]:
```python
avg_price = df.groupby("year")["average_price"].mean()

plt.figure(figsize=(10, 5))

sns.lineplot(x=avg_price.index, y=avg_price.values, marker="o")
for x, y in zip(avg_price.index, avg_price.values):
    plt.text(x, y, f"{y:.0f}",
             ha="center", va="bottom", fontsize=8)

plt.title("Average Honey Price by Year")
plt.xlabel("Year")
plt.ylabel("Average Price")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



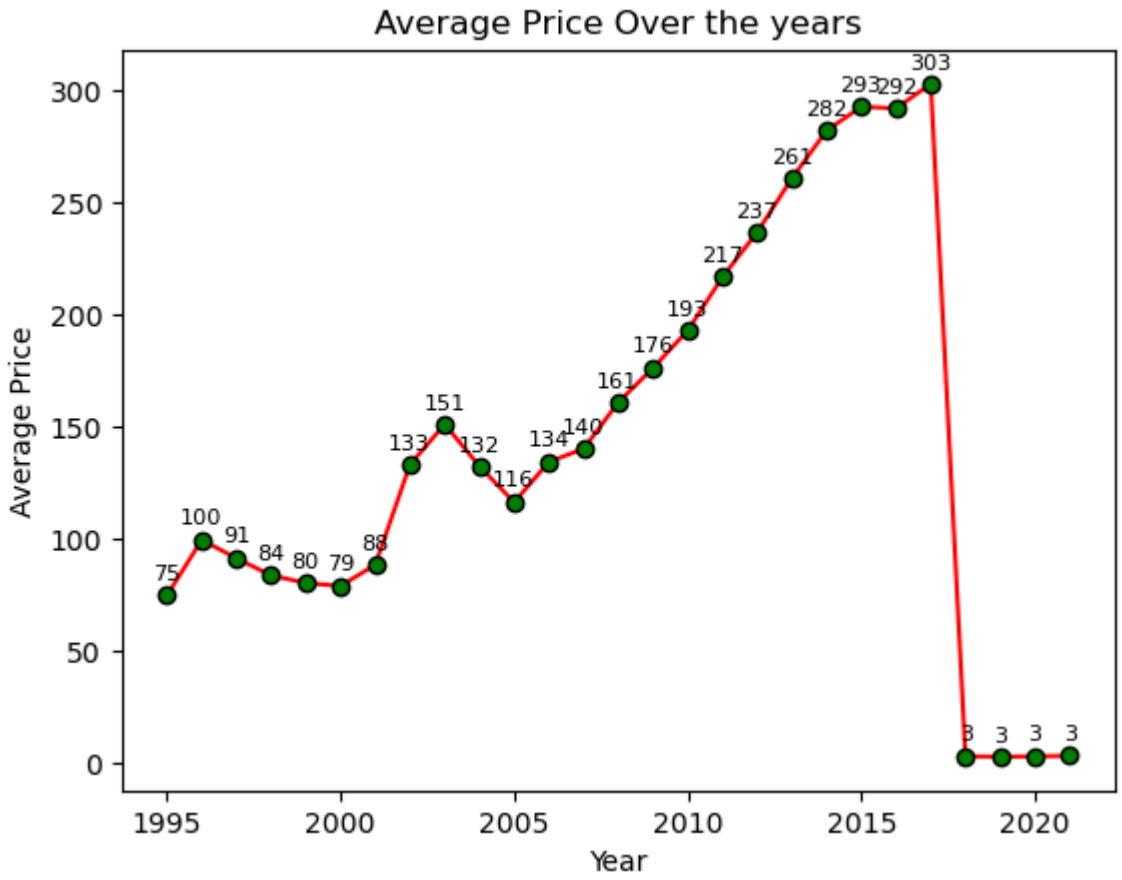
In [20]:
```
avgprice_year = df.groupby("year")["average_price"].mean()
print(avgprice_year)
```

```
year
1995     74.840909
1996     99.568182
1997     91.325581
1998     83.720930
1999     80.325581
2000     79.023256
2001     88.465116
2002    133.204545
2003    151.068182
2004    132.350000
2005    116.341463
2006    134.341463
2007    140.170732
2008    160.878049
2009    176.195122
2010    192.900000
2011    216.725000
2012    236.700000
2013    260.717949
2014    282.025000
2015    292.625000
2016    291.925000
2017    302.775000
2018      3.083500
2019      2.925500
2020      3.050000
2021      3.334250
Name: average_price, dtype: float64
```

In [21]:

```
plt.plot(avgprice_year.index,avgprice_year.values,marker = "o",color = "red",markerfacecolor = "green",markeredgecolor = "blac
for x, y in zip(avg_price.index, avg_price.values):
    plt.text(x, y +5, f"{y:.0f}",
             ha="center", va="bottom", fontsize=8)

plt.title("Average Price Over the years")
plt.xlabel("Year")
plt.ylabel("Average Price")
plt.show()
```

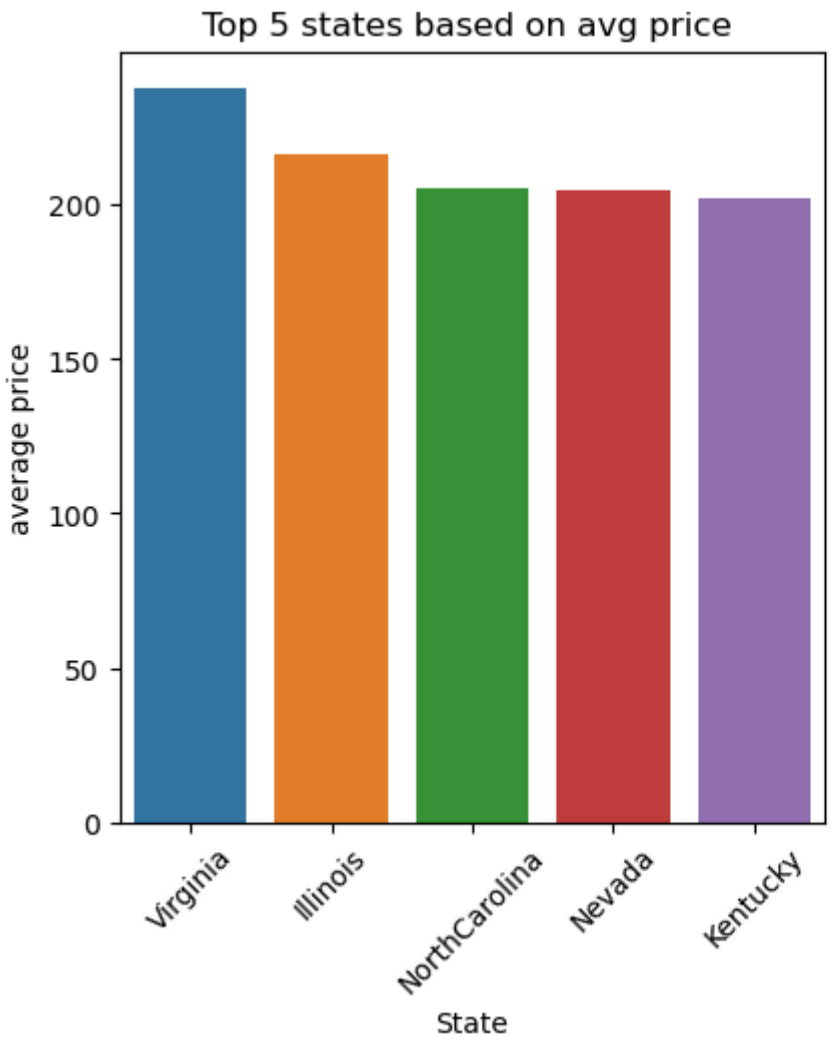


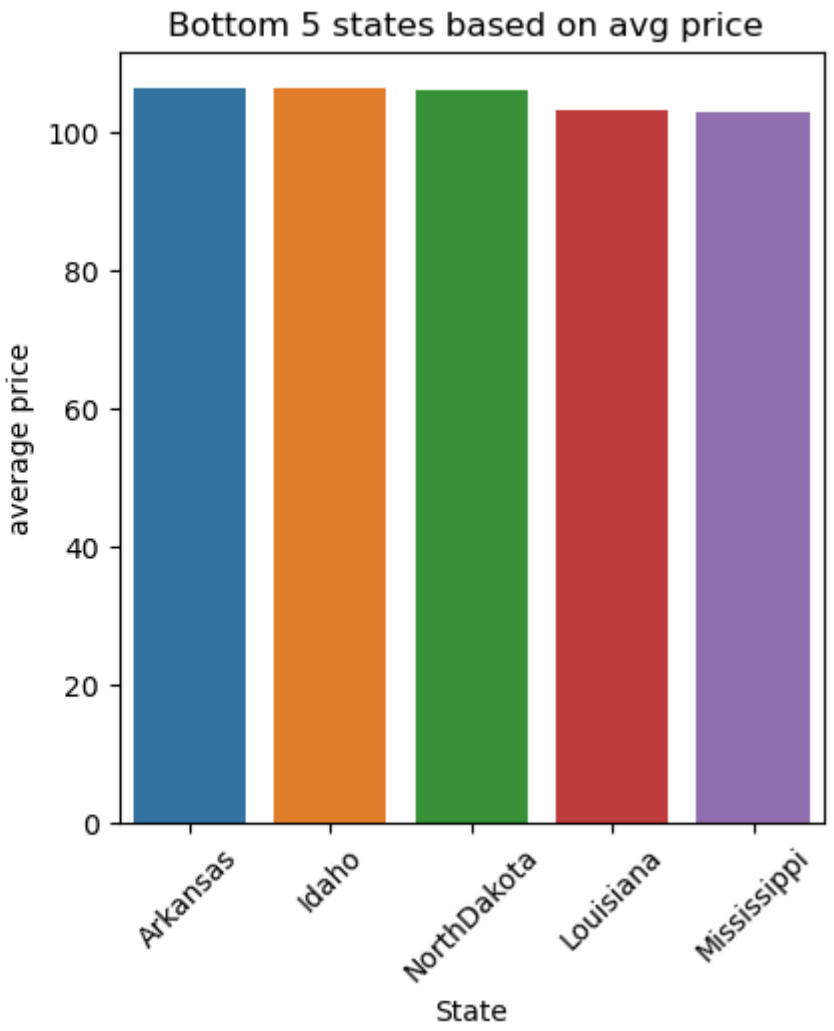
In [22]:
```
avgprice_state = df.groupby("state")["average_price"].mean()
top5_avgprice_state = avgprice_state.sort_values(ascending=False).head(5).reset_index()
print(top5_avgprice_state)
bottom5_avgprice_state = avgprice_state.sort_values(ascending=False).tail(5).reset_index()
print(bottom5_avgprice_state)
```

```
           state  average_price
0       Virginia     237.766296
1       Illinois     216.386296
2  NorthCarolina     205.192963
3         Nevada     204.533333
4       Kentucky     201.940385
         state  average_price
0     Arkansas     106.312963
1        Idaho     106.285556
2  NorthDakota     106.040000
3    Louisiana     103.074815
4  Mississippi     102.840741
```

In [23]:

```
fig = plt.subplots(1,2, figsize = (10,5))
plt.subplot(1,2,1)
sns.barplot(x="state",y="average_price",data = top5_avgprice_state ,hue = 'state' )
plt.title("Top 5 states based on avg price ")
plt.xticks(rotation = 45)
plt.xlabel("State")
plt.ylabel("average price")
plt.subplot(1,2,2)
sns.barplot(x="state",y="average_price",data = bottom5_avgprice_state ,hue = 'state' )
plt.xticks(rotation = 45)
plt.title("Bottom 5 states based on avg price ")
plt.xlabel("State")
plt.ylabel("average price")
plt.show()
```



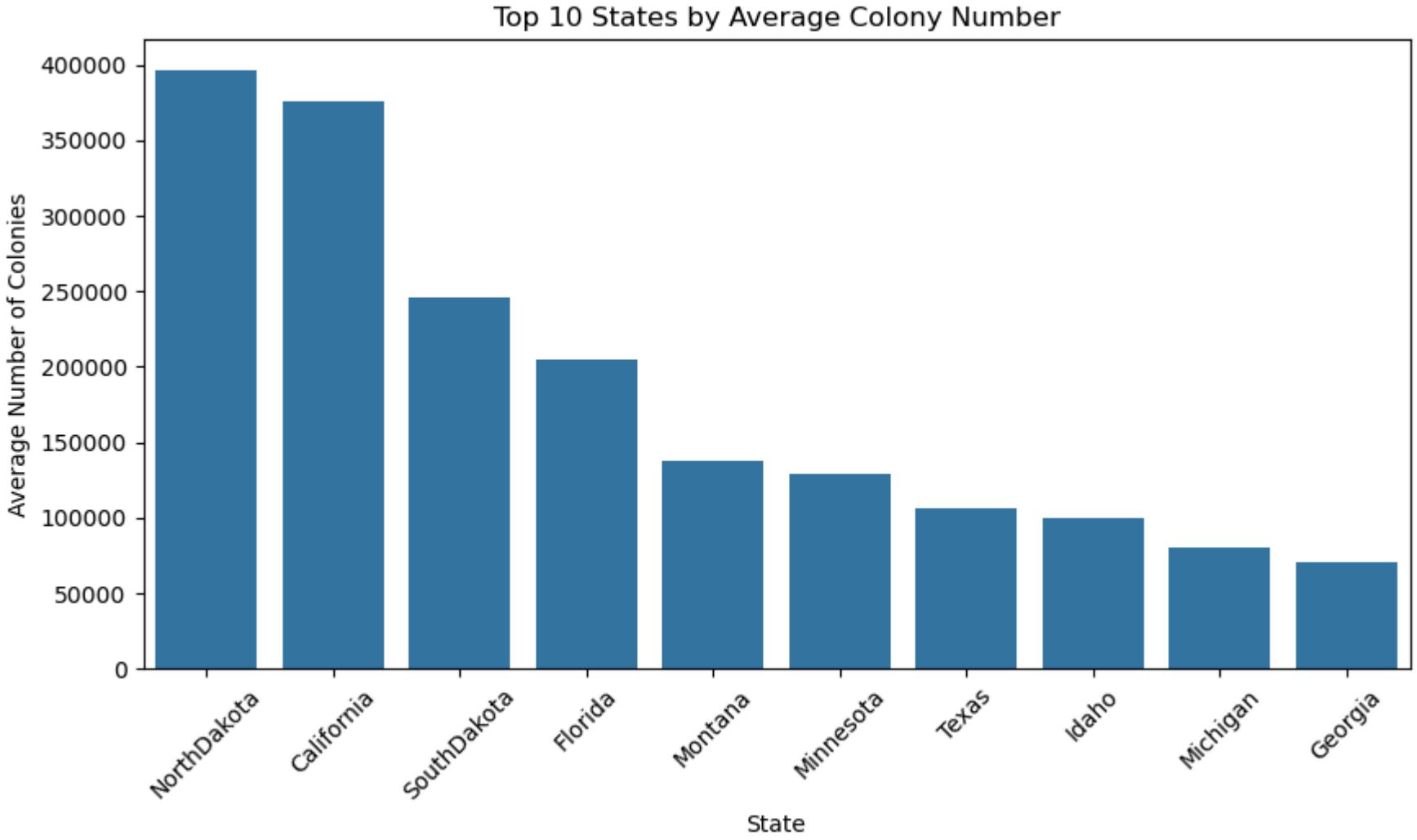


In [24]:
```
state_colonies = (
    df.groupby("state")["colonies_number"]
    .mean()
    .sort_values(ascending=False)
    .head(10)
)

plt.figure(figsize=(10, 5))
```

```
sns.barplot(x=state_colonies.index, y=state_colonies.values)

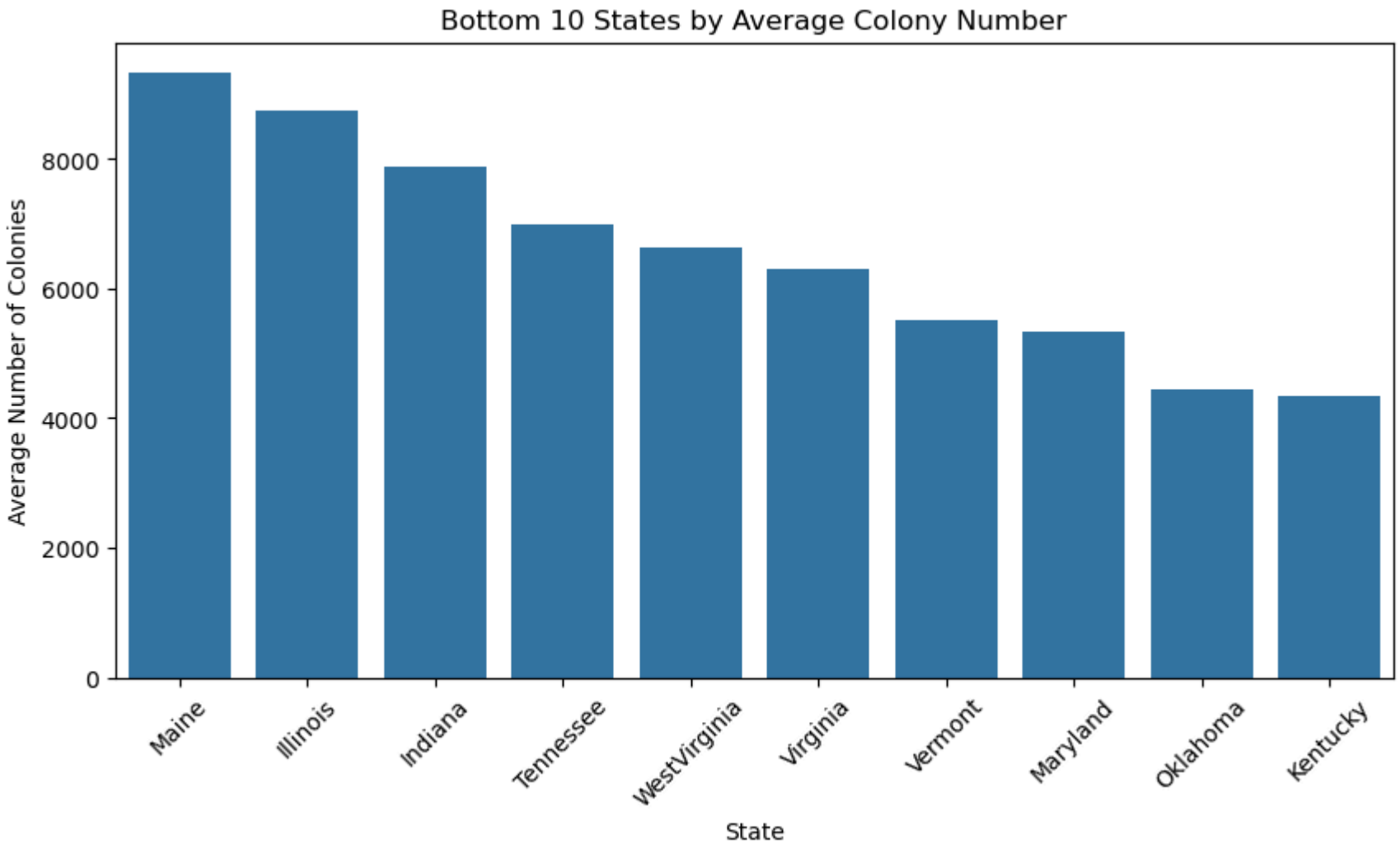
plt.title("Top 10 States by Average Colony Number")
plt.xlabel("State")
plt.ylabel("Average Number of Colonies")
plt.xticks(rotation=45)
plt.show()
```



In [25]:
```
state_colonies = (
    df.groupby("state")["colonies_number"]
    .mean()
    .sort_values(ascending=False)
    .tail(10)
)
print(state_colonies)
plt.figure(figsize=(10, 5))
sns.barplot(x=state_colonies.index, y=state_colonies.values)

plt.title("Bottom 10 States by Average Colony Number")
plt.xlabel("State")
plt.ylabel("Average Number of Colonies")
plt.xticks(rotation=45)
plt.show()
```

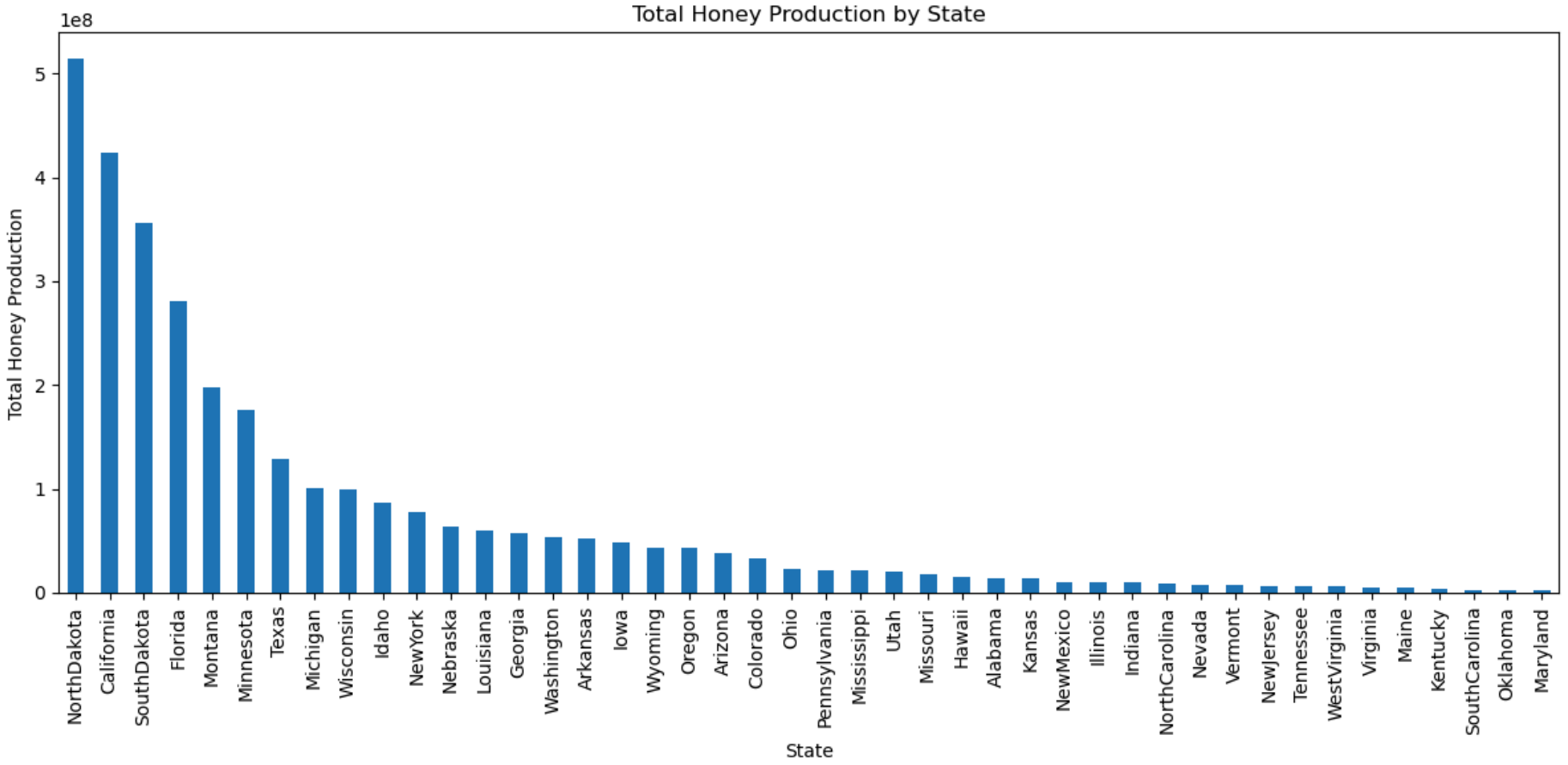
```
state
Maine           9333.333333
Illinois        8740.740741
Indiana         7888.888889
Tennessee       7000.000000
WestVirginia    6629.629630
Virginia        6296.296296
Vermont         5518.518519
Maryland        5333.333333
Oklahoma        4444.444444
Kentucky        4346.153846
Name: colonies_number, dtype: float64
```



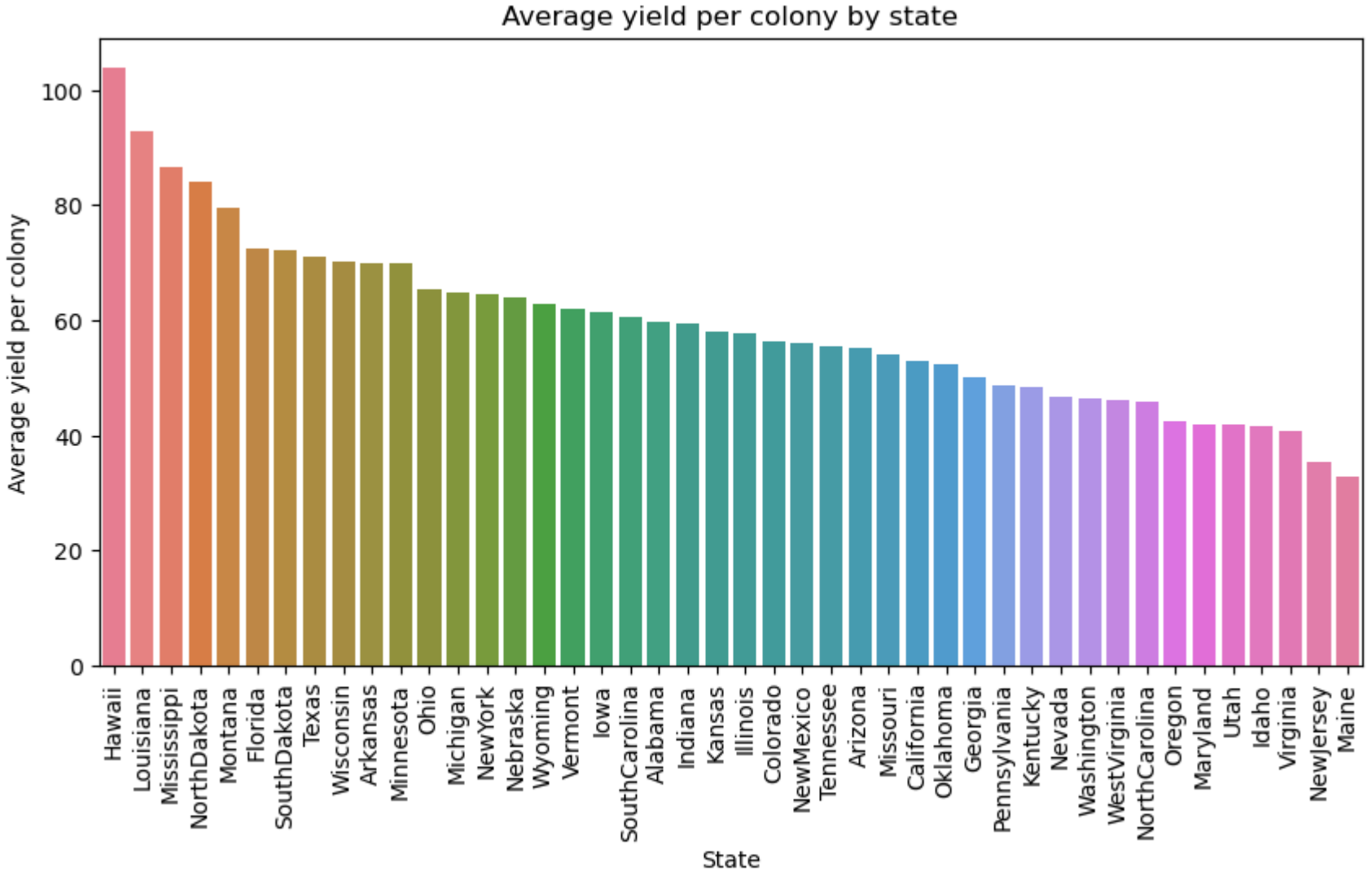
In [26]:
```
state_production =df.groupby("state")["production"].sum().sort_values(ascending = False)
state_production.plot(kind="bar", figsize=(12, 6))

plt.title("Total Honey Production by State")
plt.xlabel("State")
plt.ylabel("Total Honey Production")
plt.xticks(rotation=90)
```

```
plt.tight_layout()
plt.show()
```



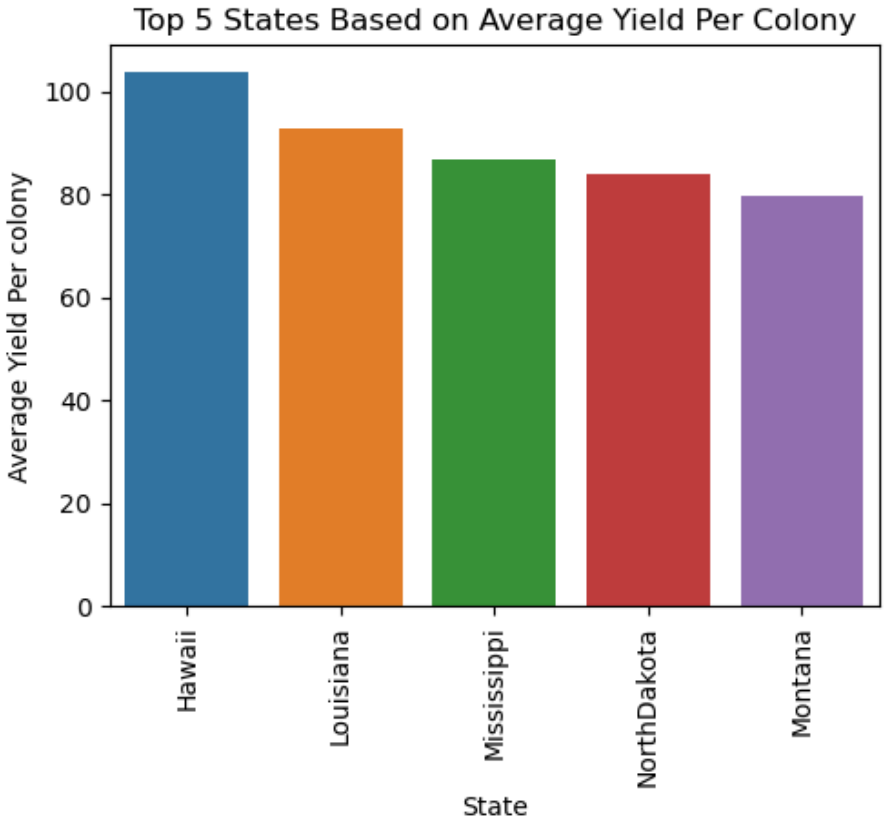
In [27]:
```
average_yield_state = df.groupby("state")["yield_per_colony"].mean().sort_values(ascending=False).reset_index()
plt.figure(figsize=(10,5))
sns.barplot(x="state", y ="yield_per_colony" ,data = average_yield_state ,hue = "state" )
plt.title("Average yield per colony by state")
plt.xlabel("State")
plt.ylabel("Average yield per colony")
plt.xticks(rotation = 90)
plt.show()
```

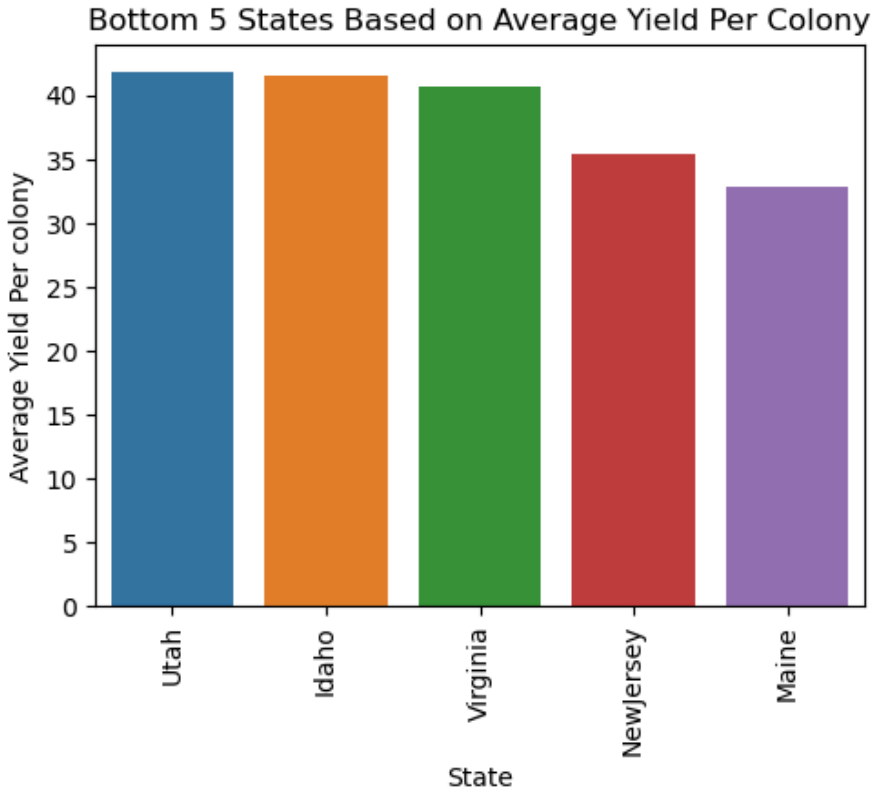


In [28]:
```
top5_state_avg_yield =average_yield_state.head(5)
bottom5_state_avg_yield =average_yield_state.tail(5)
fig = plt.subplots(1,2 ,figsize = (12,4))
plt.subplot(1,2,1)
sns.barplot(x="state", y ="yield_per_colony" ,data = top5_state_avg_yield ,hue = "state")
plt.title("Top 5 States Based on Average Yield Per Colony")
plt.xlabel("State")
```

```
plt.ylabel("Average Yield Per colony")
plt.xticks(rotation = 90)

plt.subplot(1,2,2)
sns.barplot(x="state", y ="yield_per_colony" ,data = bottom5_state_avg_yield ,hue = "state")
plt.title("Bottom 5 States Based on Average Yield Per Colony")
plt.xlabel("State")
plt.ylabel("Average Yield Per colony")
plt.xticks(rotation = 90)
plt.show()
```

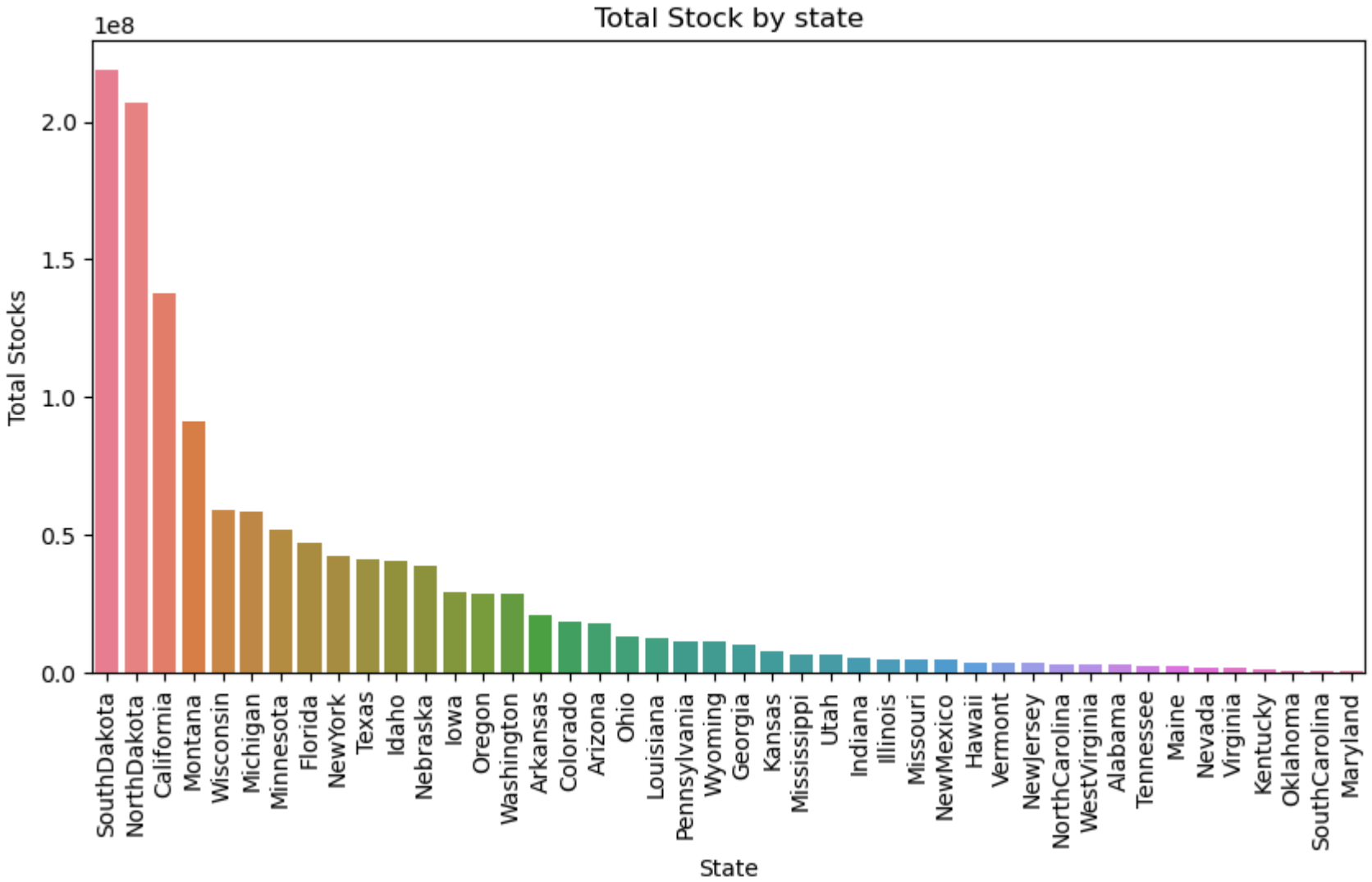




In [29]:
```
total_stock_state = df.groupby("state")["stocks"].sum().sort_values(ascending=False).reset_index()
```

In [30]:
```
plt.figure(figsize = (10,5))
sns.barplot(x = "state" , y="stocks",data =total_stock_state,hue ="state")
```

```
plt.title("Total Stock by state")
plt.xlabel("State")
plt.ylabel("Total Stocks")
plt.xticks(rotation =90)
plt.show()
```



1 . Honey Production by Year: In 1995 total honey production was approximately 21 million kg. By 2000, production had increased slightly by more than 1 million kg. However, production was decreased following years.In 2010, production dropped sharply from approximately 14 million kg to 4 million kg morethan 2 times decreased from the production in 2010 . By 2021, production had fallen further to around 2 million kg. insight: The overal trends shows significant fall in honey production over the years Possible Reasons: The decline in honey production may be caused by deforestation, which can reduce the availability of flowers and habitats for bees. Another possible reason is a reduction in production by honey-producing companies. 2. Revenue by Year : Lowest revenue: 124.19M in 1999. Highest revenue: 384.48M in 2014. Revenue comparison (2001-2002): Revenue in 2002 was approximately double the revenue in 2001, reaching 126.26M. However, honey production increased by only 1 million kg, from 17 million kg in 2001 to 18 million kg in 2002. Revenue comparison (2014-2016): Honey production remained approximately 4 million kg in 2014, 2015, and 2016, but revenue declined in 2015 and 2016 compared with 2014. insight:The bar chart shows that revenue was increasing over the years even the production was low Possible reason : due to low productoin the demand for honey in market may be increased so the prices were increased 3 Average Honey Price by Year : In 1995, the average honey price was approximately 75. The price generally increased over the years, reaching its highest value of 303 in 2017. However, in 2018, the average price dropped dramatically to approximately 3. insight : when we compare average price by year with revenu ,the revenue doesnt effect even the average price decreased Possible Reasons:Average honey prices generally increased until 2017, followed by an unusual decline in 2018 that requires further investigation. 4.Top and Bottom 5 states based on avg price : TOP 5 states are Virginia,Illinois,NorthCarolina,Nevada,Kentucky Bottom 5 states are Arkansas,Idaho,NorthDakota,Louisiana ,Mississippi insight : when we see top and bottom 5 states average prices the difference is huge from 237.77 to 102.84 Possible Reasons: Honey prices vary significantly across states. 5.Top and Bottom 10 States by Average Colony Number : Top 10 States by Average Colony Number :Northdakota ,californina, southdakota,florida,montana,minnesota,texas,idaho,michigan and georgia has highest average colony number insight: In the top 10 state the highest averge colony numbers having northdakota there is significant decrease number of colony in every state when compared to northdakota Bottom 10 States by Average Colony Number :Maine,Illinois,Indiana,Tennessee,WestVirginia ,Virginia,Vermont,Maryland ,Oklahoma and Kentucky insight:In bottom state the lowest averge colony numbers having Oklahoma and Kentucky Possible Reasons:North Dakota has the highest average colony number, while Oklahoma and Kentucky are among the lowest. 6.Total Honey Production by State : The highetst and lowest honey production states are North California and Maryland respectively insight : The chart shows significant decrease over the state 7.Average yield per colony by state: The highest state of average yield per colony is hawaii The lowest state of average yield per colony is maine the middle of the states are sightly same average yield per colony it shows yield per colony is not effect on production of honey so fewer number of colonies effect on the total production Possible Reasons:Hawaii has the highest average yield per colony, whereas Maine has the lowest. 8.Top and bottom 5 States Based on Average Yield Per Colony : Top 5 States Based on Average Yield Per Colony : hawaii,Louisiana,Mississippi,Northdakota and Montana Bottom 5 States Based on Average Yield Per Colony : Utah ,Idaho,Virgina,Newjersey and Maine insight: the both plots tells average yield per colony .the bottom states are yielding per colony half of the top state yielding per

colony Possible Reasons:Both colony numbers and productivity per colony play important roles in determining total honey production. 9.Total Stock by state : The maximum stocks left in southdakota and Northdakota The minimum stocks left in maryland insight : southdakota and Northdakota having maximum stocks andmaryl and having minimum stocks Possible Reasons:Remaining stock levels differ significantly across states

Final Business Recommendation: Further analysis should focus on understanding the relationship between colony numbers, honey yield, production volume, average selling prices, revenue, and remaining stock. This could help identify states with strong production potential, improve productivity, and support better inventory and pricing decisions.